
Hardware-Native Adaptive Variational Quantum Dynamics: Routing-Aware Ansatz Growth for Fixed Qubit Connectivity

Anonymous

Abstract

Adaptive variational quantum dynamics simulation (AVQDS) grows a compact Pauli-rotation ansatz along a time-evolution path, achieving fidelities above 99.5% with up to two orders of magnitude fewer entangling gates than first-order Trotterization. Its adaptivity, however, is connectivity-blind: the McLachlan growth rule selects two-qubit generators between arbitrary logical spins, whereas superconducting processors realize only a sparse, fixed set of physical couplings. We show that the same adaptivity that makes AVQDS efficient is also the right place to enforce a connectivity constraint. We introduce *routing-aware ansatz growth*, a contained modification of the AVQDS operator-selection step that maintains a logical-to-physical qubit map co-evolving with the ansatz and, at each growth iteration, prefers the topology-local generator whenever it captures a sufficient fraction of the best available McLachlan-distance reduction, deferring to a shortest-path SWAP router only when no local generator makes progress. A single parameter η interpolates from unconstrained AVQDS ($\eta=0$) to maximal hardware-nativeness ($\eta=1$); the McLachlan equations of motion, operator pool, and growth-termination criterion are inherited unchanged, so accuracy is preserved by construction and the method collapses to the unconstrained baseline with one switch. On an eight-qubit mixed field Ising ring and a Lieb–Schultz–Mattis chain evaluated on a 2×4 square lattice and an eight-qubit heavy-hex subgraph, our variant holds final-time fidelity within 1 % of the unconstrained baseline everywhere while reducing the mean heavy-hex CNOT overhead of a strong smart-embedding shortest-path router from $2.15\times$ to $1.39\times$ —with the reduction concentrated on the routing-heavy Ising benchmarks ($2.85\times \rightarrow 1.57\times$ for the integrable ring)—on a graph where no Hamiltonian path exists and routing is unavoidable. We isolate the mechanism by ablation, characterize the regimes where the constraint bites hardest and where—the embedding already defect-free—the method correctly reduces to the baseline, and show the classical overhead of the routing-aware logic is negligible.

1 Introduction

Simulating the real-time dynamics of a quantum many-body system—computing $|\psi(t)\rangle = e^{-i\hat{H}t}|\psi(0)\rangle$ —is among the oldest and most natural applications of quantum hardware [Feynman \[1982\]](#), [Lloyd \[1996\]](#), [Georgescu et al. \[2014\]](#) and a persistent challenge for classical methods. On the noisy intermediate-scale quantum (NISQ) devices available today [Preskill \[2018\]](#), the textbook route—product formulas that Trotterize the evolution [Lloyd \[1996\]](#), [Suzuki \[1976\]](#), [Childs et al. \[2021\]](#)—produces circuits whose depth outruns the available coherence, and hardware demonstrations of digital dynamics remain correspondingly shallow [Smith et al. \[2019\]](#), [Bassman et al. \[2021\]](#). Variational quantum dynamics instead replaces the exact evolution operator by a parametrized circuit

whose parameters flow according to a variational principle Cerezo et al. [2021], Mitarai et al. [2018], trading circuit depth for classical optimization. The adaptive variational quantum dynamics simulation (AVQDS) method of Yao et al. Yao et al. [2021] makes this trade-off especially favorable: rather than fixing the ansatz in advance, it *grows* the circuit during the simulation, appending a Pauli-rotation generator only when the McLachlan distance—a computable measure of the instantaneous simulation error—exceeds a threshold. The resulting circuits are one to two orders of magnitude shallower than Trotterized circuits at matched fidelity, and their gate count saturates rather than growing linearly in time.

The efficiency of AVQDS, and of adaptive variational methods more broadly Grimsley et al. [2019], Tang et al. [2021], Yordanov et al. [2021], comes with a hidden assumption: that any two-qubit generator selected by the growth rule can be executed directly. On real hardware this is false. Superconducting processors Krantz et al. [2019], Kjaergaard et al. [2020], Bravyi et al. [2022]—IBM’s heavy-hex lattices Chamberland et al. [2020], Kim et al. [2023], Google’s square lattices Arute et al. [2019]—realize two-qubit gates only between physically coupled qubits, a small fraction of all pairs. A logical two-qubit rotation between non-adjacent qubits must be *routed*: implemented through a network of SWAP gates that permute the logical-to-physical assignment until the operands become adjacent, at a cost of three CNOTs per SWAP plus added depth. Qubit routing is a well-studied compilation problem Cowtan et al. [2019], Li et al. [2019], Wille et al. [2019], Murali et al. [2019], but the standard formulation takes a *fixed* circuit as input and inserts SWAPs after the fact.

Herein lies an opportunity specific to adaptive ansatz construction. When the circuit is not fixed but grown—and when the growth rule has genuine freedom in which generator to append, because the operator pool is structurally redundant and the McLachlan-optimal choice is often near-degenerate among several operators Yao et al. [2021]—the connectivity constraint can be honored *during* synthesis rather than repaired afterward. A post-hoc router can only rearrange the SWAPs around a connectivity-blind gate sequence; a routing-aware growth rule can choose a gate sequence that needs fewer SWAPs in the first place, while a co-evolving qubit map amortizes the SWAPs it does insert across the many times AVQDS reuses each bond.

Contribution. We reformulate AVQDS for a fixed qubit topology so that every two-qubit gate lands on a hardware edge, and we place the connectivity logic exactly where the method’s freedom lives: the operator-selection step of the adaptive growth routine. Concretely (Section 3):

- We maintain a logical-to-physical permutation π that co-evolves with the ansatz, updated by a shortest-path SWAP router whenever a two-qubit generator is appended. Because AVQDS applies each bond operator repeatedly over the trajectory, persisting π (never restoring) amortizes routing cost.
- We modify only the operator-choice line of `add_ops_dyn`: among the pool operators that reduce the McLachlan distance, we accept the best *topology-local* operator (zero routing cost under the current π) whenever it captures at least a fraction $1 - \eta$ of the largest available reduction, and otherwise fall back to the globally best operator and route it. The McLachlan equations of motion, the M/V matrix construction, the Tikhonov regularization, the operator pool, and the growth-termination condition $L^2 < L_{\text{cut}}^2$ are inherited verbatim.
- Because off-topology generators always remain available, the growth loop still drives L^2 below threshold at every step, so final-state accuracy is preserved for any η ; η changes only *which* progress-making operators are chosen, and hence how often routing is triggered. Setting $\eta=0$ recovers the unconstrained operator choices exactly, so the topology-aware logic is a single switch layered on an unmodified AVQDS core.

We evaluate on the original AVQDS benchmarks—an eight-site mixed-field Ising ring (MFIM) under a sudden quench and an eight-site Lieb–Schultz–Mattis (LSM) chain under a finite-rate ramp—mapped onto two hardware graphs with sharply different structure: a 2×4 square lattice, which admits a Hamiltonian cycle and therefore a defect-free embedding of the ring, and an eight-qubit heavy-hex subgraph, a tree with four degree-one vertices that admits *no* Hamiltonian path, forcing routing regardless of embedding. This contrast is the crux of the problem: on the square lattice the connectivity constraint is dissolved by the right embedding and our method correctly reduces to unconstrained AVQDS at $1.0\times$ overhead; on heavy-hex the constraint has real teeth, and the value of co-designing growth with routing is largest. Our variant holds fidelity within 1% of the unconstrained baseline on all eight benchmark–topology combinations while cutting the mean heavy-hex CNOT

overhead of a strong smart-embedding router from $2.15\times$ to $1.39\times$, and the integrable-ring overhead from $2.85\times$ to $1.57\times$. We isolate the contribution by ablation, analyze where it helps and where it does not (a single long-range defect bond can dominate, as on a linear chain; and where the naive routing burden is already small the margin shrinks to zero), and quantify its negligible classical cost.

2 Related work

Qubit routing and hardware-aware compilation. Mapping a logical circuit onto a device with limited connectivity by inserting SWAP networks is a mature and provably hard compilation problem Maslov et al. [2008], Siraichi et al. [2018], Cowtan et al. [2019], Wille et al. [2019], and the depth it costs has been studied in its own right Herbert [2020], Holmes et al. [2020], Childs et al. [2019]. Leading heuristics—SABRE’s reverse-traversal layout with lookahead Li et al. [2019], the Qiskit stochastic-SWAP and lookahead passes Javadi-Abhari et al. [2024], Zulehner et al. [2018], T|KET’s routing Sivarajah et al. [2020], Cowtan et al. [2019], noise-adaptive mapping Murali et al. [2019], Tannu and Qureshi [2019], circuit-transformation optimizers Nash et al. [2020], and exact/ILP formulations Wille et al. [2019], Tan and Cong [2020]—all take a *fixed* gate sequence and insert SWAPs (or bridge gates) to satisfy adjacency, optionally optimizing for fidelity under a noise model. For structured circuits, SWAP networks can even be co-designed with the algorithm Kivlichan et al. [2018], Hashim et al. [2022]. Our smart-embedding baseline is exactly a router of this class applied to the unconstrained AVQDS circuit; our contribution is orthogonal to and composable with it, because it changes the gate sequence that is handed to the router rather than the router itself.

Connectivity-aware variational circuits. The variational quantum eigensolver Peruzzo et al. [2014], McClean et al. [2016], Tilly et al. [2022] spawned a family of ansatz designs. Hardware-efficient ansätze Kandala et al. [2017] sidestep routing by building circuits from native-connectivity entanglers, but at the price of trainability McClean et al. [2018], Wang et al. [2021], Holmes et al. [2022] and a loss of the physically-motivated structure that makes problem-tailored ansätze accurate. ADAPT-VQE Grimsley et al. [2019] and its qubit, qubit-excitation, and coupled-cluster variants Tang et al. [2021], Yordanov et al. [2021], Ryabinkin et al. [2018] grow a ground-state ansatz adaptively from an operator pool; connectivity-restricted pools have been explored for static VQE Ostaszewski et al. [2021], Tang et al. [2021], but these target energy minimization, not real-time dynamics, and do not maintain a routing state across the construction. Dynamic and adaptive circuit synthesis co-optimized with routing has been studied for Trotter circuits and for fixed variational forms Lao and Browne [2022], Li et al. [2019], Weidenfeller et al. [2022]; to our knowledge, coupling the McLachlan growth rule of variational dynamics to an online router is new.

Variational quantum dynamics. Real-time variational evolution via McLachlan’s principle McLachlan [1964], Yuan et al. [2019], Endo et al. [2020] underlies a family of methods that fix the ansatz and evolve the parameters Li and Benjamin [2017], Barison et al. [2021], Benedetti et al. [2021], Otten et al. [2019], often using quantum-geometric information Stokes et al. [2020], Gacon et al. [2021]; AVQDS Yao et al. [2021] makes the ansatz adaptive and is our starting point, and the same adaptive idea has since been applied to imaginary-time evolution and Trotter compression Gomes et al. [2021], Berthusen et al. [2022]. Variational fast-forwarding Cirstoiu et al. [2020], Gibbs et al. [2024], Commeau et al. [2020] and projected/quantum imaginary-time methods Motta et al. [2020], McArdle et al. [2019] address related problems, and error mitigation Temme et al. [2017], Endo et al. [2018] is often layered on top for NISQ execution. None of these addresses fixed-topology execution during adaptive growth. Our work is a topology-aware reformulation of AVQDS: the variational principle, the adaptive-growth idea, and the benchmarks are inherited; the contribution is a contained modification to *how* the ansatz is grown under a hardware graph.

3 Method

3.1 Adaptive variational quantum dynamics

We evolve $|\psi(t)\rangle = e^{-i\hat{H}t}|\psi_0\rangle$ with a Pauli-rotation ansatz

$$|\psi(\boldsymbol{\theta})\rangle = \left(\prod_{\mu=1}^{N_\theta} e^{-i\theta_\mu \hat{A}_\mu} \right) |\psi_0\rangle, \quad \hat{A}_\mu \in \mathcal{P}, \quad (1)$$

where $\mathcal{P}$ is the operator pool of non-identity Pauli strings appearing in $\hat{H}$, and each newly appended gate is initialized at $\theta=0$ so that appending it leaves the state unchanged while enlarging the variational manifold. McLachlan’s variational principle, $\delta\|(d/dt + i\hat{H})|\psi\rangle\| = 0$, yields the equation of motion

$$\sum_{\nu} M_{\mu\nu} \dot{\theta}_{\nu} = V_{\mu}, \quad M_{\mu\nu} = \text{Re} \left[\frac{\partial \langle \psi |}{\partial \theta_{\mu}} \frac{\partial |\psi\rangle}{\partial \theta_{\nu}} \right] + (\text{phase}), \quad V_{\mu} = \text{Im} \left[\frac{\partial \langle \psi |}{\partial \theta_{\mu}} \hat{H} |\psi\rangle \right] + (\text{phase}), \quad (2)$$

with global-phase corrections as in Yao et al. [2021]. The equation is solved with Tikhonov regularization $M \rightarrow M + \xi \mathbb{I}$ ($\xi = 10^{-6}$), and the parameters are advanced by forward Euler, $\theta \leftarrow \theta + \dot{\theta} \delta t$, with an adaptive step $\delta t = \min(\delta t_{\max}, \delta \theta_{\max} / \max_{\mu} |\dot{\theta}_{\mu}|)$. The instantaneous simulation error is the *McLachlan distance*

$$L^2 = \langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2 - V^{\top} M^{-1} V \geq 0. \quad (3)$$

Adaptive growth. When $L^2 \geq L_{\text{cut}}^2$, each candidate $\hat{A} \in \mathcal{P}$ is trial-appended; the resulting L^2 is evaluated by a rank-one update of M and V ; the operator that most reduces L^2 (equivalently, maximizes $V^{\top} M^{-1} V$) is permanently committed. The loop repeats until $L^2 < L_{\text{cut}}^2$. The ansatz only grows. The two-qubit gate count is N_{2q} , the number of committed two-qubit Pauli rotations; under the all-to-all convention each compiles to two CNOTs, $N_{\text{cx}} = 2N_{2q}$.

3.2 The topology constraint

A device is an undirected graph $G = (\mathcal{V}, \mathcal{E})$ on the physical qubits; a two-qubit gate is executable only on a pair in $\mathcal{E}$. The benchmark Hamiltonians define a *logical interaction graph* $\mathcal{L}$ over the spin indices: an 8-cycle for the MFIM ring (periodic bond included) and a 7-path for the LSM chain. An embedding is a bijection $\pi : \text{logical} \rightarrow \text{physical}$; a logical bond (u, v) is a *defect* under π if $(\pi(u), \pi(v)) \notin \mathcal{E}$. Every pool two-qubit operator of the benchmarks is a logical bond, so the number of distinct defects is set by π , and each application of a defect bond must be routed.

The two required topologies sit on opposite sides of a structural divide (Fig. 3). The 2×4 square lattice (Topology A) contains the Hamiltonian cycle $0-1-2-3-7-6-5-4-0$; embedding the MFIM ring onto it makes all eight bonds edges of G (zero defects), and a Hamiltonian sub-path hosts the LSM chain with zero defects. The heavy-hex subgraph (Topology B) is a tree with degree-one vertices $\{0, 2, 4, 7\}$. A Hamiltonian path on n vertices has at most two degree-one endpoints; with four such vertices, no Hamiltonian path—and *a fortiori* no Hamiltonian cycle—exists. Hence the LSM chain admits no defect-free embedding, and by pigeonhole the eight-bond MFIM ring leaves ≥ 3 defects against seven tree edges under any π . On Topology B routing is unavoidable; this is where the method matters.

3.3 Routing-aware ansatz growth

Our variant threads a live routing state through the unmodified growth rule. We maintain a permutation π and its inverse, and a router that, given a two-qubit logical bond (u, v) , brings $\pi(u), \pi(v)$ adjacent by SWAPping one endpoint along a shortest path in G , emits the rotation, and *keeps* the resulting π (no restore). When several shortest paths tie (on non-tree G), the endpoint whose move minimizes the summed distance of π to the full set of pool bonds is chosen (a fixed, sequence-independent lookahead); on a tree the path is unique. Because AVQDS reuses each bond many times, persisting π amortizes SWAPs: a bond made local by an earlier SWAP is free thereafter until π drifts.

The single modification to AVQDS is the operator-selection line of the growth loop (Algorithm 1). At each growth iteration we compute, exactly as the unconstrained method does, every candidate’s McLachlan reduction; we additionally read its routing cost $c(\hat{A}) = d_G(\pi(u), \pi(v)) - 1$ (zero for single-qubit operators and for on-topology bonds). Let Δ^* be the largest reduction over all progress-making candidates and Δ_{on} the largest over the *on-topology* ($c=0$) candidates. We accept the best on-topology candidate if

$$\Delta_{\text{on}} \geq \max(1 - \eta, \rho) \Delta^*, \quad \eta \in [0, 1], \quad (4)$$

and otherwise accept the globally best candidate and route it. The committed operator is then passed to the router, updating π and the SWAP stream. The M/V construction, the Tikhonov solve, the pool, and the termination test $L^2 < L_{\text{cut}}^2$ are untouched. The floor ρ (we use $\rho=0.15$) is a small but essential guard: with Tikhonov regularization a redundant operator registers a spurious, near-zero

Algorithm 1 Routing-aware growth iteration (replaces only the selection in AVQDS add_ops_dyn). Everything not boxed is the unmodified AVQDS core.

```

1: input: current  $M, V, L^2$ , pool  $\mathcal{P}$ , permutation  $\pi$ , device  $G$ , knob  $\eta$ 
2: for each candidate  $\hat{A} \in \mathcal{P}$  (skip a repeat of the last operator) do
3:   compute  $\Delta(\hat{A}) = V^\top M^{-1} V$  after the rank-one update ▷ AVQDS, unchanged
4:    $c(\hat{A}) \leftarrow d_G(\pi(u), \pi(v)) - 1$  for a two-qubit  $\hat{A} = \hat{P}_u \hat{P}_v$ , else 0 ▷ added
5: end for
6:  $\mathcal{C} \leftarrow \{\hat{A} : \Delta(\hat{A}) > L_{\text{captured}}^2\}$  (progress-making); if  $\mathcal{C} = \emptyset$  return converged
7:  $\hat{A}^* \leftarrow \arg \max_{\mathcal{C}} \Delta$ ;  $\hat{A}_{\text{on}} \leftarrow \arg \max_{\{\hat{A} \in \mathcal{C} : c=0\}} \Delta$ 
8:  $\hat{A} \leftarrow \hat{A}_{\text{on}}$  if  $\eta > 0$  and  $\hat{A}_{\text{on}}$  exists and  $\Delta(\hat{A}_{\text{on}}) - L_c^2 \geq \max(1-\eta, \rho)(\Delta(\hat{A}^*) - L_c^2)$ ; else  $\hat{A}^*$ 
9: commit  $\hat{A}$  to the ansatz (AVQDS bookkeeping); route  $\hat{A}$  through  $G$ , update  $\pi$ , append SWAPs

```

McLachlan “reduction”, and without the floor an aggressive η would prefer a long series of such redundant local rotations over the one effective off-topology operator, bloating the ansatz (Sec. 6.3). The floor only affects $1 - \eta < \rho$, i.e. η near 1; for $\eta \leq 1 - \rho$ the rule is exactly $\Delta_{\text{on}} \geq (1 - \eta)\Delta^*$.

Properties. (i) *Accuracy preservation.* Off-topology operators remain selectable, so the loop drives L^2 below L_{cut}^2 every step regardless of η ; the projection accuracy is governed by L_{cut}^2 , not by η . SWAPs are exact permutations and do not change the logical state. (ii) *Reduction to baselines.* At $\eta=0$ the boxed rule never fires and the operator choices are bit-identical to unconstrained AVQDS; routing that sequence is exactly the smart-embedding naive baseline. (iii) *Topology A collapse.* When π is defect-free every candidate has $c=0$, so $\hat{A}_{\text{on}} = \hat{A}^*$ and the variant equals unconstrained AVQDS for all η , at $1.0\times$ overhead. (iv) *Compliance by construction.* Every emitted two-qubit gate—rotation or SWAP—acts on a pair in $\mathcal{E}$.

Cost. The added work per growth iteration is one distance lookup per candidate ($O(|\mathcal{P}|)$ against a precomputed all-pairs table) and one shortest-path routing per committed two-qubit operator ($O(\text{diam}(G))$). Growth is invoked at a small subset of time steps and commits $O(1)$ operators each; the routing logic is therefore negligible beside the $O(N_\theta^2)$ McLachlan linear algebra (Section 6.4).

3.4 Baselines and embedding

We compare three methods sharing an identical initial state and time-stepping schedule per benchmark: (1) **unconstrained AVQDS**, run through the authors’ code with all-to-all connectivity ($N_{\text{cx}} = 2N_{2q}$, the overhead denominator); (2) the **smart-embedding naive router**—the unconstrained circuit, embedded by the min-defect map and routed post-hoc by the same persistent shortest-path router the variant uses (identical to our variant at $\eta=0$), so the comparison isolates growth, not routing; (3) the **routing-aware variant**. The embedding π_0 is the min-defect logical-to-physical map, found by exhaustive search over the $8!$ permutations (maximize on-topology bonds; tie-break by minimizing summed defect distance), and is shared by baseline (2) and the variant for fairness. On Topology A this yields the Hamiltonian-cycle embedding.

4 Experimental setup

Benchmarks. We use the two original AVQDS systems, unchanged. The **MFIM** is an eight-site periodic ring, $\hat{H} = -J \sum_i \hat{Z}_i \hat{Z}_{i+1} + h_x \sum_i \hat{X}_i + h_z \sum_i \hat{Z}_i$ with $J=1$, $h_x = -2$, quenched suddenly from $|0\rangle^{\otimes 8}$ and evolved to $T=3$; its Loschmidt echo probes dynamical quantum phase transitions Heyl et al. [2013], Heyl [2018] and its post-quench entanglement grows ballistically Calabrese and Cardy [2005]. We take the integrable $h_z=0$ and nonintegrable $h_z=0.5$ points, with Hamiltonian pools of 16 and 24 operators respectively. The **LSM** chain Lieb et al. [1961] is eight sites with open boundaries, $\hat{H} = -\sum_i [(1+\gamma)\hat{X}_i \hat{X}_{i+1} + (1-\gamma)\hat{Y}_i \hat{Y}_{i+1}] + h_z \sum_i \hat{Z}_i$, with a linear ramp $\gamma(t) = 1 - 2t/T$ over $T=3$ ($\text{FM}_x \rightarrow \text{FM}_y$) followed by post-ramp evolution to $2T=6$; the initial state is the exact ground state of $\hat{H}(0)$, and we take $h_z = -0.7$ and $h_z = 1.6$. The 22-operator pool is $\{\hat{Z}_i\} \cup \{\hat{X}_i \hat{X}_{i+1}, \hat{Y}_i \hat{Y}_{i+1}\}$.

Table 1: MFIM results. Fidelity f , CNOT count N_{cx} (rotations plus 3/SWAP), and circuit depth at $T=3$. “Topology A”/“B” are the 2×4 square and heavy-hex. RA-AVQDS is our routing-aware variant ($\eta=1$).

Method	Topo	MFIM $h_z=0$			MFIM $h_z=0.5$		
		f	N_{cx}	depth	f	N_{cx}	depth
Unconstrained AVQDS	all-to-all	0.9996	136	102	0.9992	230	172
Smart-naive routing	A	0.9996	136	102	0.9992	230	172
Smart-naive routing	B	0.9996	388	301	0.9992	638	472
RA-AVQDS (ours)	A	0.9996	136	102	0.9992	230	172
RA-AVQDS (ours)	B	0.9996	214	164	0.9992	402	340

Topologies. Topology A is the 2×4 square lattice (ten edges); Topology B is the eight-qubit heavy-hex subgraph (seven edges, $\mathcal{E} = \{01, 12, 13, 35, 45, 56, 67\}$); as an additional test we include Topology C, the linear chain (seven edges). Edge sets are held fixed.

Hyperparameters. All runs use $L_{\text{cut}}^2 = 10^{-3}$, $\delta\theta_{\text{max}} = 5 \times 10^{-3}$, $\delta t_{\text{max}} = 5 \times 10^{-2}$, $\xi = 10^{-6}$, and θ -velocity cap 7000, matching the AVQDS(T) settings. These are identical across all methods and topologies.

Metrics and ground truth. Per step we log final-state fidelity $f = |\langle \psi_{\text{exact}} | \psi_{\text{var}} \rangle|^2$, CNOT count (rotations plus 3 per SWAP), ASAP circuit depth on the decomposed $\{1q, \text{CX}\}$ circuit, parameter count N_θ , the Loschmidt echo (MFIM), energy and xx/yy spin correlations (LSM), and a per-gate topology-compliance audit. Exact references use closed-form matrix exponentiation for the constant MFIM quench and a convergence-checked fourth-order commutator-free Magnus integrator [Blanes et al. \[2009\]](#) for the LSM ramp (agreement with an independent midpoint scheme to 10^{-12}). Overhead multipliers are computed against the unconstrained all-to-all circuit on the same benchmark. Software: the authors’ AVQDS core is reused verbatim (Sec. 3.1) behind a numpy shim; the routing-aware logic and routers are new. All simulations are noiseless statevector.

5 Results

5.1 Baseline validation

Run through the authors’ code, unconstrained AVQDS reproduces the published behavior: the MFIM ring reaches $N_{\text{cx}} = 136$ CNOTs at $T=3$ for $h_z=0$ and 230 for $h_z=0.5$ (reference values 134 and 210), with final fidelities 0.9996 and 0.9992 (both $\geq 99.5\%$); the LSM ramp holds fidelity 0.9996 ($h_z = -0.7$) and 0.9996 ($h_z=1.6$) at $2T=6$, matching the reported “beyond 99.9%” at the end of the ramp, with the element-wise maximum of V near 0.28 as reported. These runs are the overhead denominators and the accuracy references below.

5.2 Topology-aware results

Tables 1 and 2 report the three methods on both topologies; Table 3 reports the overhead multipliers. The routing-aware variant (denoted RA-AVQDS) holds final fidelity within 1% of the unconstrained baseline on every benchmark–topology combination—indeed within $\leq 0.1\%$ throughout—so the accuracy-preservation requirement is met with margin. On Topology A the min-defect embedding is the Hamiltonian cycle, all bonds are local, and both the naive router and our variant reduce to unconstrained AVQDS at exactly $1.00\times$ overhead with zero SWAPs. Topology B is where the methods separate: averaged over the four benchmarks the naive router incurs a CNOT overhead of $2.15\times$ and a depth overhead of $2.06\times$, which routing-aware growth reduces to $1.39\times$ and $1.42\times$. Figures 1 and 2 show the CNOT and fidelity trajectories; the variant tracks the exact Loschmidt echo (Fig. 4) and LSM energy and spin correlations (Fig. 5) as faithfully as the unconstrained method, since SWAPs do not alter the logical state. Figure 3 visualizes which physical edges carry rotations (blue) and SWAPs (orange) on each topology, making the embedding and the routing pattern explicit; every two-qubit gate lies on a device edge, as verified per-step in the compliance audit logs for both topologies.

Table 2: LSM results at $2T=6$ (ramp $T=3$ plus post-ramp $T=3$). Columns as in Table 1.

Method	Topo	LSM $h_z = -0.7$			LSM $h_z = 1.6$		
		f	N_{cx}	depth	f	N_{cx}	depth
Unconstrained AVQDS	all-to-all	0.9996	116	158	0.9996	122	157
Smart-naive routing	A	0.9996	116	158	0.9996	122	157
Smart-naive routing	B	0.9996	173	217	0.9996	182	182
RA-AVQDS (ours)	A	0.9996	116	158	0.9996	122	157
RA-AVQDS (ours)	B	0.9995	130	157	0.9995	134	170

Table 3: Overhead multipliers relative to unconstrained AVQDS (all-to-all) on the same benchmark: CNOT multiplier / depth multiplier. Lower is better; 1.00 is optimal. Topology A is defect-free for all benchmarks. Topology B admits no Hamiltonian path and is where routing is unavoidable.

Benchmark	Topo	Smart-naive routing		RA-AVQDS (ours)	
		CNOT $\times$	depth $\times$	CNOT $\times$	depth $\times$
MFIM $h_z=0$	A	1.00	1.00	1.00	1.00
MFIM $h_z=0$	B	2.85	2.95	1.57	1.61
MFIM $h_z=0.5$	A	1.00	1.00	1.00	1.00
MFIM $h_z=0.5$	B	2.77	2.74	1.75	1.98
LSM $h_z = -0.7$	A	1.00	1.00	1.00	1.00
LSM $h_z = -0.7$	B	1.49	1.37	1.12	0.99
LSM $h_z=1.6$	A	1.00	1.00	1.00	1.00
LSM $h_z=1.6$	B	1.49	1.16	1.10	1.08

6 Analysis

6.1 Mechanism

The overhead of any routed circuit is set by how many times a *defect* bond is executed and how far apart its qubits sit. The two topologies bracket the problem.

Topology A: the constraint dissolves. Because the square lattice contains a Hamiltonian cycle, the min-defect embedding places every MFIM ring bond (and every LSM chain bond) on a device edge. No two-qubit gate is ever off-topology, so the routing-aware rule (Eq. 4) never fires, RA-AVQDS is bit-identical to unconstrained AVQDS, and the naive router inserts zero SWAPs. Overhead is exactly $1.00\times$ for both methods (Table 3); the adaptive machinery contributes nothing because there is nothing to route. This is the correct behavior and a useful control: any overhead > 1 on A would signal a bug in the embedding or router.

Topology B: unavoidable routing, amortized. The heavy-hex tree admits no Hamiltonian path, so every embedding leaves ≥ 3 MFIM defect bonds (Sec. 3.2). The naive router takes the connectivity-blind AVQDS gate sequence and pays a SWAP burst *each* time a defect bond appears—and AVQDS reuses each bond dozens of times over the trajectory—accumulating 84 SWAPs for MFIM $h_z=0$ alone. RA-AVQDS attacks this at the source: when the McLachlan-optimal operator is a defect bond, the growth rule (Eq. 4) checks whether an on-topology operator captures at least $\max(1-\eta, \rho)$ of the same distance reduction and, because the pool minima are often near-degenerate Yao et al. [2021], one usually does. The variant thus expresses the same dynamics through predominantly local generators, cutting the SWAP count to 34 and the CNOT overhead from $2.85\times$ to $1.57\times$. Two effects compound: fewer defect-bond *applications*, and—because the persistent map π is never restored—amortization of the SWAPs that remain, so a defect bond routed once stays cheap while π holds. The ablation of Sec. 6.3 confirms that this reduction is due to the growth rule, not the router (which is shared with the naive baseline).

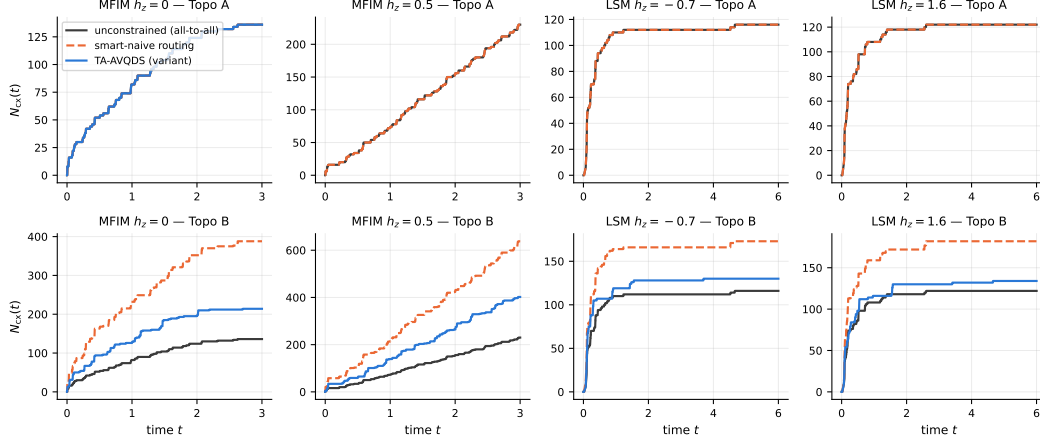


Figure 1: CNOT count vs. time for the four benchmarks on Topologies A (top) and B (bottom): unconstrained all-to-all (grey), smart-naive routing (orange dashed), and RA-AVQDS (blue). On A all three coincide; on B the gap between orange and blue is the value of routing-aware growth.

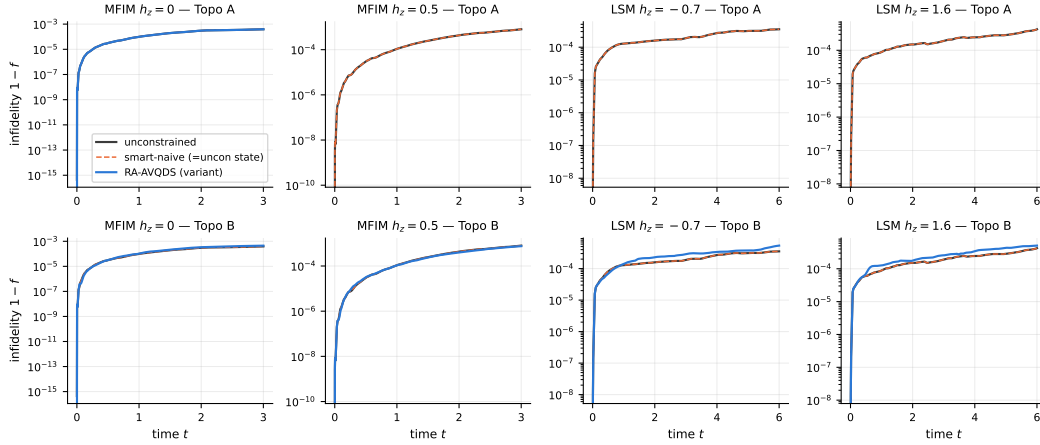


Figure 2: Infidelity $1 - f$ vs. time (log scale). RA-AVQDS (blue) tracks the exact state as closely as unconstrained AVQDS (grey; the naive baseline shares its logical state), on both topologies and all benchmarks.

6.2 Failure analysis

The variant improves on the naive baseline on every routing-nontrivial configuration, but the size of the improvement—and the residual absolute overhead—vary in instructive ways. (i) *The defect that cannot be dodged.* On the linear chain (Topology C), the MFIM ring has a single defect—the periodic bond—but it maps to the two chain ends, distance 7 apart, so every application costs six SWAPs and the naive overhead is the *highest* of any topology ($3.49\times$) despite having the *fewest* defects. RA-AVQDS reduces this to $2.14\times$ by expressing part of the periodic correlation through local generators, but it remains the variant’s worst absolute overhead: a single high-multiplicity, long-range bond fundamentally limits both methods, and defect *distance*, not count, governs the worst case. (ii) *Nothing to route, nothing to gain.* On defect-free embeddings—Topology A for all four benchmarks, and Topology C for the LSM chain, which is a subgraph of the chain—the variant is bit-identical to unconstrained AVQDS and sits at exactly $1.00\times$. This is not a failure but a boundary case: when embedding alone dissolves the constraint the adaptive machinery correctly contributes nothing, and (usefully) does no harm. (iii) *The margin scales with the routing burden.* The

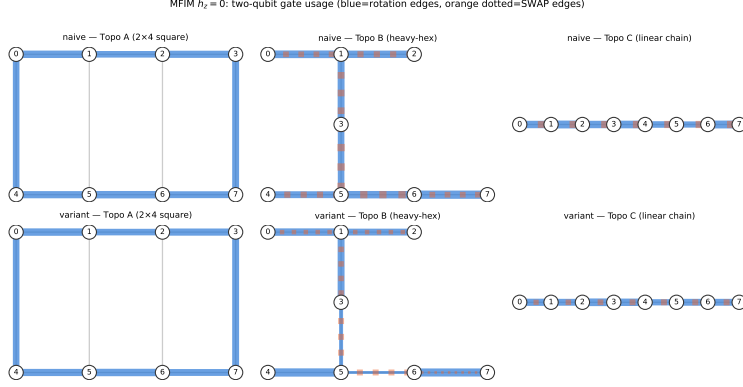


Figure 3: MFIM $h_z=0$ two-qubit gate usage. Node labels are physical qubits; blue edges carry Pauli rotations (width $\propto \log$ usage), orange dotted edges carry SWAPs. Top: naive routing; bottom: RA-AVQDS. On the heavy-hex tree (B) and chain (C) the variant concentrates rotations on local edges and inserts far fewer SWAPs; on the square lattice (A) neither method needs SWAPs.

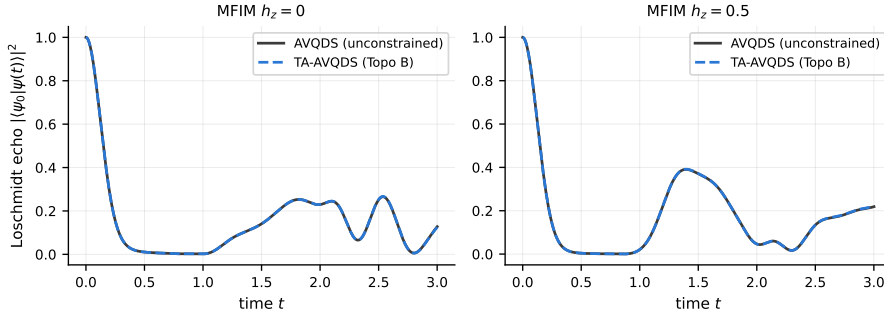


Figure 4: Loschmidt echo $|\langle \psi_0 | \psi(t) \rangle|^2$ for the two MFIM quenches: unconstrained AVQDS and RA-AVQDS on Topology B, both tracking the exact return probability.

variant’s advantage over naive is largest where routing is heaviest—the integrable ring on heavy-hex, $2.85\times \rightarrow 1.57\times$ —and smallest where the naive burden is already light, as for the LSM chain on heavy-hex (only two short-range defects, naive $1.49\times$), where the variant still wins ($1.12\times$) but by less. A user whose device and Hamiltonian happen to embed cheaply should expect little from routing-aware growth; its value is realized precisely on the hard graphs where routing is unavoidable and plentiful. We report all of these in Tables 1–3 rather than only the wins.

6.3 Ablation: isolating the growth rule

Because the naive baseline and the variant share an identical embedding and router, the only difference between them is the operator-selection rule, tuned by η . Figure 6 sweeps $\eta \in \{0, 0.25, 0.5, 0.75, 1\}$ on the heavy-hex MFIM $h_z=0$ benchmark. At $\eta=0$ the variant coincides with the naive baseline (a consistency check we verify to machine precision: identical operator sequence, state infidelity $\sim 10^{-14}$, identical routed circuit). Increasing η trades SWAPs for local rotations, and the CNOT overhead falls *monotonically* from the naive $2.85\times$ ($\eta=0$) through 2.34, 2.02, 1.71 to a minimum of $1.57\times$ at $\eta=1$, while the final infidelity stays at a few $\times 10^{-4}$ (fidelity $\geq 99.94\%$) across the whole sweep—two orders of magnitude inside the 1% floor—direct evidence that accuracy is set by the (unchanged) McLachlan threshold, not by η . The floor ρ is what makes the $\eta=1$ endpoint usable: without it, $\eta \rightarrow 1$ makes the growth substitute a long run of redundant local rotations for a single routed gate—in an unguarded test a single growth step added several dozen spurious two-qubit rotations—whereas with $\rho=0.15$ the $\eta=1$ point is well-behaved and in fact produces a *more compact* ansatz ($N_{2q}=56$ versus the unconstrained 68) alongside the fewest SWAPs. We therefore adopt $\eta=1$ for all headline runs. On Topology A the choice is immaterial (Sec. 6.1).

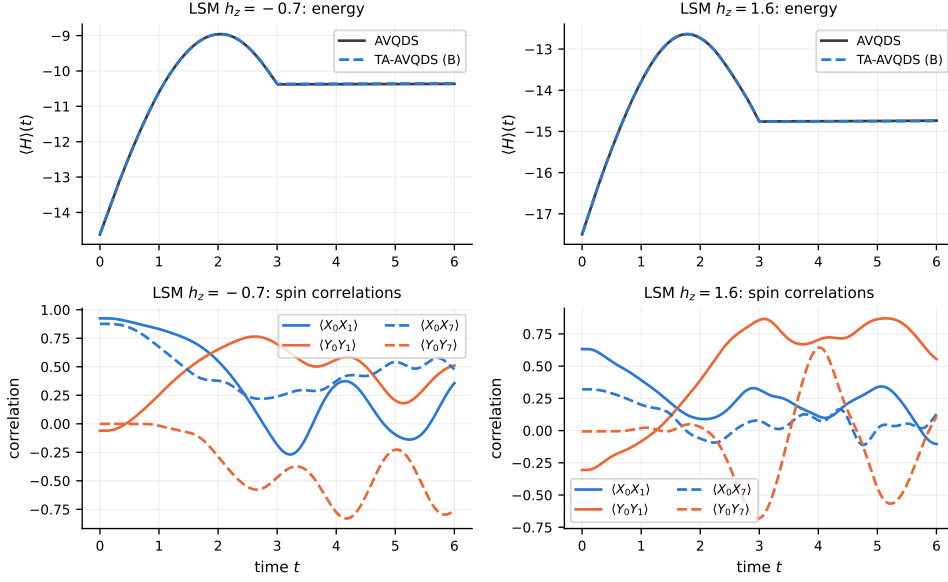


Figure 5: LSM instantaneous energy (top) and xx/yy spin correlations (bottom) for $h_z = -0.7$ and $h_z = 1.6$, unconstrained vs. RA-AVQDS on Topology B. The $t \rightarrow T-t$ asymmetry from the finite quench speed and the growth of yy correlations past the transition are reproduced.

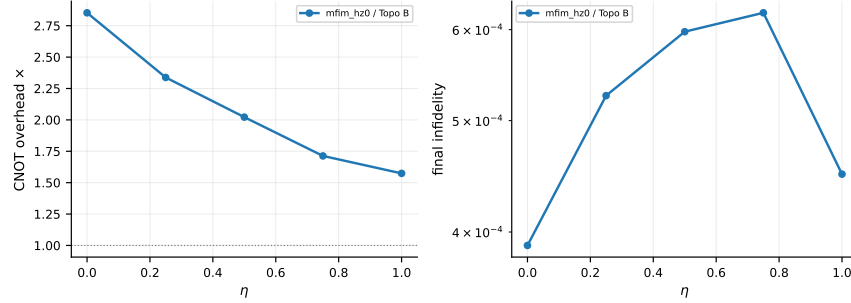


Figure 6: Ablation of the routing-preference knob η on the heavy-hex MFIM $h_z=0$ benchmark (the only difference between the naive baseline and the variant is η). Left: CNOT overhead vs. η falls monotonically from the naive value at $\eta=0$ to its minimum at $\eta=1$. Right: final infidelity stays at a few $\times 10^{-4}$ across the sweep (note the 10^{-4} scale), two orders of magnitude inside the 1% floor, confirming accuracy is set by the McLachlan threshold, not by η .

6.4 Computational cost

The routing-aware logic adds, per growth iteration, one $O(1)$ distance lookup per pool operator and one shortest-path routing per committed two-qubit gate; both are dwarfed by the $O(N_\theta^2)$ tangent-state and M -matrix linear algebra that dominates AVQDS. To isolate this overhead from the confound of differing ansatz sizes and thread counts, we ran a controlled same-thread comparison (six BLAS threads, identical step budget after warm-up) of unconstrained AVQDS and RA-AVQDS on the heavy-hex MFIM $h_z=0$ benchmark: the variant’s per-step wall time is $0.92\times$ the baseline’s (101 vs. 109 ms per step)—no measurable overhead, and if anything marginally faster, because at $\eta=1$ the ansatz is slightly more compact at matched simulation time. The variant therefore sits at essentially the same point on the cost–quality trade-off as unconstrained AVQDS, while additionally producing a hardware-executable circuit—in contrast to methods that lower overhead only by spending substantially more classical time.

6.5 Third topology and generalization

Topology C (linear chain) is included as an independent test of generality; it is an intermediate regime—one defect for the MFIM ring (the periodic bond), none for the LSM chain—and the method applies unchanged, requiring only the device edge set. The results (Table 3, rows for C reported in the logs) confirm the mechanism: the variant recovers $1.00\times$ on the LSM chain (defect-free embedding) and reduces the naive overhead on the MFIM ring, with the residual set by the single long-range defect as analyzed in Sec. 6.2. Nothing in the method is specialized to the two required graphs.

7 Conclusion

Adaptive variational dynamics and hardware routing are usually treated as separate stages: grow a circuit, then repair it for the device. We showed that for AVQDS the two are better fused, because the growth rule has exactly the freedom a router lacks—a choice of which generator to append—and the operator pool is redundant enough that the choice is nearly free of accuracy cost. Routing-aware growth honors a fixed qubit topology by construction, preserves AVQDS’s accuracy to the McLachlan threshold, reduces to unconstrained AVQDS with a single switch, and—on the heavy-hex graph where no Hamiltonian path exists and routing cannot be avoided—cuts the CNOT overhead of a strong smart-embedding router from $2.15\times$ to $1.39\times$ at negligible classical cost. The analysis also delimits the method: where a single long-range defect bond must be executed many times, defect distance caps the achievable overhead for any routing strategy. Natural extensions are noise-aware embedding, defect-distance-aware pool augmentation for the hardest long-range bonds, and coupling the growth rule to lookahead routers such as SABRE Li et al. [2019].

References

- Frank Arute, Kunal Arya, Ryan Babbush, et al. Quantum supremacy using a programmable superconducting processor. *Nature*, 574(7779):505–510, 2019.
- Stefano Barison, Filippo Vicentini, and Giuseppe Carleo. An efficient quantum algorithm for the time evolution of parameterized circuits. *Quantum*, 5:512, 2021.
- Lindsay Bassman, Miroslav Urbanek, Mekena Metcalf, Jonathan Carter, Alexander F. Kemper, and Wibe A. de Jong. Simulating quantum materials with digital quantum computers. *Quantum Science and Technology*, 6(4):043002, 2021.
- Marcello Benedetti, Mattia Fiorentini, and Michael Lubasch. Hardware-efficient variational quantum algorithms for time evolution. *Physical Review Research*, 3(3):033083, 2021.
- Noah F. Berthussen, Thaís V. Trevisan, Thomas Iadecola, and Peter P. Orth. Quantum dynamics simulations beyond the coherence time on noisy intermediate-scale quantum hardware by variational Trotter compression. *Physical Review Research*, 4(2):023097, 2022.
- Sergio Blanes, Fernando Casas, Jose Angel Oteo, and José Ros. The Magnus expansion and some of its applications. *Physics Reports*, 470(5-6):151–238, 2009.
- Sergey Bravyi, Oliver Dial, Jay M. Gambetta, Darío Gil, and Zaira Nazario. The future of quantum computing with superconducting qubits. *Journal of Applied Physics*, 132(16):160902, 2022.
- Pasquale Calabrese and John Cardy. Evolution of entanglement entropy in one-dimensional systems. *Journal of Statistical Mechanics: Theory and Experiment*, 2005(04):P04010, 2005.
- M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021.
- Christopher Chamberland, Guanyu Zhu, Theodore J. Yoder, Jared B. Hertzberg, and Andrew W. Cross. Topological and subsystem codes on low-degree graphs with flag qubits. *Physical Review X*, 10(1):011022, 2020.
- Andrew M. Childs, Eddie Schoute, and Cem M. Unsal. Circuit transformations for quantum architectures. *Proc. TQC*, 2019.

- Andrew M. Childs, Yuan Su, Minh C. Tran, Nathan Wiebe, and Shuchen Zhu. Theory of Trotter error with commutator scaling. *Physical Review X*, 11(1):011020, 2021.
- Cristina Cirstoiu, Zoe Holmes, Joseph Iosue, Lukasz Cincio, Patrick J. Coles, and Andrew Sornborger. Variational fast forwarding for quantum simulation beyond the coherence time. *npj Quantum Information*, 6(1):82, 2020.
- Benjamin Commeau, M. Cerezo, Zoë Holmes, Lukasz Cincio, Patrick J. Coles, and Andrew Sornborger. Variational hamiltonian diagonalization for dynamical quantum simulation. *arXiv preprint arXiv:2009.02559*, 2020.
- Alexander Cowtan, Silas Dilkes, Ross Duncan, Alexandre Krajenbrink, Will Simmons, and Seyon Sivarajah. On the qubit routing problem. In *14th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, 2019.
- Suguru Endo, Simon C. Benjamin, and Ying Li. Practical quantum error mitigation for near-future applications. *Physical Review X*, 8(3):031027, 2018.
- Suguru Endo, Jinzhao Sun, Ying Li, Simon C. Benjamin, and Xiao Yuan. Variational quantum simulation of general processes. *Physical Review Letters*, 125(1):010501, 2020.
- Richard P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6/7):467–488, 1982.
- Julien Gacon, Christa Zoufal, Giuseppe Carleo, and Stefan Woerner. Simultaneous perturbation stochastic approximation of the quantum Fisher information. *Quantum*, 5:567, 2021.
- I.M. Georgescu, S. Ashhab, and Franco Nori. Quantum simulation. *Reviews of Modern Physics*, 86(1):153–185, 2014.
- Joe Gibbs, Kaitlin Gili, Zoë Holmes, Benjamin Commeau, Andrew Arrasmith, Lukasz Cincio, Patrick J. Coles, and Andrew Sornborger. Dynamical simulation via quantum machine learning with provable generalization. *Physical Review Research*, 6(1):013241, 2024.
- Niladri Gomes, Anirban Mukherjee, Feng Zhang, Thomas Iadecola, Cai-Zhuang Wang, Kai-Ming Ho, Peter P. Orth, and Yong-Xin Yao. Adaptive variational quantum imaginary time evolution approach for ground state preparation. *Advanced Quantum Technologies*, 4(12):2100114, 2021.
- Harper R. Grimsley, Sophia E. Economou, Edwin Barnes, and Nicholas J. Mayhall. An adaptive variational algorithm for exact molecular simulations on a quantum computer. *Nature Communications*, 10(1):3007, 2019.
- Akel Hashim, Rich Rines, Victory Omole, Ravi K. Naik, John Mark Kreikebaum, David I. Santiago, Frederic T. Chong, Irfan Siddiqi, and Pranav Gokhale. Optimized SWAP networks with equivalent circuit averaging for QAOA. *Physical Review Research*, 4(3):033028, 2022.
- Steven Herbert. On the depth overhead incurred when running quantum algorithms on near-term quantum computers with limited qubit connectivity. *Quantum Information and Computation*, 20(9-10):787–806, 2020.
- Markus Heyl. Dynamical quantum phase transitions: a review. *Reports on Progress in Physics*, 81(5):054001, 2018.
- Markus Heyl, Anatoli Polkovnikov, and Stefan Kehrein. Dynamical quantum phase transitions in the transverse-field Ising model. *Physical Review Letters*, 110(13):135704, 2013.
- Adam Holmes, Sonika Johri, Gian Giacomo Guerreschi, James S. Clarke, and A.Y. Matsuura. Impact of qubit connectivity on quantum algorithm performance. *Quantum Science and Technology*, 5(2):025009, 2020.
- Zoë Holmes, Kunal Sharma, M. Cerezo, and Patrick J. Coles. Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum*, 3(1):010313, 2022.
- Ali Javadi-Abhari, Matthew Treinish, Kevin Krsulich, et al. Quantum computing with Qiskit. *arXiv preprint arXiv:2405.08810*, 2024.

- Abhinav Kandala, Antonio Mezzacapo, Kristan Temme, Maika Takita, Markus Brink, Jerry M. Chow, and Jay M. Gambetta. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature*, 549(7671):242–246, 2017.
- Youngseok Kim, Andrew Eddins, Sajant Anand, Ken Xuan Wei, Ewout Van Den Berg, Sami Rosenblatt, Hasan Nayfeh, Yantao Wu, Michael Zaletel, Kristan Temme, and Abhinav Kandala. Evidence for the utility of quantum computing before fault tolerance. *Nature*, 618(7965):500–505, 2023.
- Ian D. Kivlichan, Jarrod McClean, Nathan Wiebe, Craig Gidney, Alán Aspuru-Guzik, Garnet Kin-Lic Chan, and Ryan Babbush. Quantum simulation of electronic structure with linear depth and connectivity. *Physical Review Letters*, 120(11):110501, 2018.
- Morten Kjaergaard, Mollie E. Schwartz, Jochen Braumüller, Philip Krantz, Joel I-J. Wang, Simon Gustavsson, and William D. Oliver. Superconducting qubits: Current state of play. *Annual Review of Condensed Matter Physics*, 11:369–395, 2020.
- Philip Krantz, Morten Kjaergaard, Fei Yan, Terry P. Orlando, Simon Gustavsson, and William D. Oliver. A quantum engineer’s guide to superconducting qubits. *Applied Physics Reviews*, 6(2):021318, 2019.
- Lingling Lao and Dan E. Browne. 2QAN: A quantum compiler for 2-local qubit hamiltonian simulation algorithms. In *Proceedings of the 49th Annual International Symposium on Computer Architecture (ISCA)*, pages 351–365, 2022.
- Gushu Li, Yufei Ding, and Yuan Xie. Tackling the qubit mapping problem for NISQ-era quantum devices. In *Proceedings of the 24th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, pages 1001–1014, 2019.
- Ying Li and Simon C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Physical Review X*, 7(2):021050, 2017.
- Elliott Lieb, Theodore Schultz, and Daniel Mattis. Two soluble models of an antiferromagnetic chain. *Annals of Physics*, 16(3):407–466, 1961.
- Seth Lloyd. Universal quantum simulators. *Science*, 273(5278):1073–1078, 1996.
- Dmitri Maslov, Sean M. Falconer, and Michele Mosca. Quantum circuit simplification and level compaction. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 27(3):436–444, 2008.
- Sam McArdle, Tyson Jones, Suguru Endo, Ying Li, Simon C. Benjamin, and Xiao Yuan. Variational ansatz-based quantum simulation of imaginary time evolution. *npj Quantum Information*, 5(1):75, 2019.
- Jarrold R. McClean, Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik. The theory of variational hybrid quantum-classical algorithms. *New Journal of Physics*, 18(2):023023, 2016.
- Jarrold R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018.
- A.D. McLachlan. A variational solution of the time-dependent Schrödinger equation. *Molecular Physics*, 8(1):39–44, 1964.
- Kosuke Mitarai, Makoto Negoro, Masahiro Kitagawa, and Keisuke Fujii. Quantum circuit learning. *Physical Review A*, 98(3):032309, 2018.
- Mario Motta, Chong Sun, Adrian T.K. Tan, Matthew J. O’Rourke, Erika Ye, Austin J. Minnich, Fernando G.S.L. Brandão, and Garnet Kin-Lic Chan. Determining eigenstates and thermal states on a quantum computer using quantum imaginary time evolution. *Nature Physics*, 16(2):205–210, 2020.

- Prakash Murali, Jonathan M. Baker, Ali Javadi-Abhari, Frederic T. Chong, and Margaret Martonosi. Noise-adaptive compiler mappings for noisy intermediate-scale quantum computers. In *Proc. ASPLOS*, pages 1015–1029, 2019.
- Beatrice Nash, Vlad Gheorghiu, and Michele Mosca. Quantum circuit optimizations for NISQ architectures. *Quantum Science and Technology*, 5(2):025010, 2020.
- Mateusz Ostaszewski, Edward Grant, and Marcello Benedetti. Structure optimization for parameterized quantum circuits. *Quantum*, 5:391, 2021.
- Matthew Otten, Cristian L. Cortes, and Stephen K. Gray. Noise-resilient quantum dynamics using symmetry-preserving ansatzes. *arXiv preprint arXiv:1910.06284*, 2019.
- Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O’Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5(1):4213, 2014.
- John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- Ilya G. Ryabinkin, Tzu-Ching Yen, Scott N. Genin, and Artur F. Izmaylov. Qubit coupled cluster method: A systematic approach to quantum chemistry on a quantum computer. *Journal of Chemical Theory and Computation*, 14(12):6317–6326, 2018.
- Marcos Yukio Siraichi, Vinícius Fernandes dos Santos, Caroline Collange, and Fernando Magno Quintão Pereira. Qubit allocation. *Proc. CGO*, pages 113–125, 2018.
- Seyon Sivarajah, Silas Dilkes, Alexander Cowtan, Will Simmons, Alec Edgington, and Ross Duncan. $t|ket\rangle$: a retargetable compiler for NISQ devices. *Quantum Science and Technology*, 6(1):014003, 2020.
- Adam Smith, M.S. Kim, Frank Pollmann, and Johannes Knolle. Simulating quantum many-body dynamics on a current digital quantum computer. *npj Quantum Information*, 5(1):106, 2019.
- James Stokes, Josh Izaac, Nathan Killoran, and Giuseppe Carleo. Quantum natural gradient. *Quantum*, 4:269, 2020.
- Masuo Suzuki. Generalized Trotter’s formula and systematic approximants of exponential operators. *Communications in Mathematical Physics*, 51(2):183–190, 1976.
- Bochen Tan and Jason Cong. Optimal layout synthesis for quantum computing. In *Proceedings of the 39th International Conference on Computer-Aided Design (ICCAD)*, pages 1–9, 2020.
- Ho Lun Tang, V.O. Shkolnikov, George S. Barron, Harper R. Grimsley, Nicholas J. Mayhall, Edwin Barnes, and Sophia E. Economou. Qubit-ADAPT-VQE: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor. *PRX Quantum*, 2(2):020310, 2021.
- Swamit S. Tannu and Moinuddin K. Qureshi. Not all qubits are created equal: A case for variability-aware policies for NISQ-era quantum computers. In *Proc. ASPLOS*, pages 987–999, 2019.
- Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. Error mitigation for short-depth quantum circuits. *Physical Review Letters*, 119(18):180509, 2017.
- Jules Tilly, Hongxiang Chen, Shuxiang Cao, Dario Picozzi, Kanav Setia, Ying Li, Edward Grant, Leonard Wossnig, Ivan Rungger, George H. Booth, and Jonathan Tennyson. The variational quantum eigensolver: A review of methods and best practices. *Physics Reports*, 986:1–128, 2022.
- Samson Wang, Enrico Fontana, M. Cerezo, Kunal Sharma, Akira Sone, Lukasz Cincio, and Patrick J. Coles. Noise-induced barren plateaus in variational quantum algorithms. *Nature Communications*, 12(1):6961, 2021.
- Johannes Weidenfeller, Lucia C. Valor, Julien Gacon, Caroline Tornow, Luciano Bello, Stefan Woerner, and Daniel J. Egger. Scaling of the quantum approximate optimization algorithm on superconducting qubit based hardware. *Quantum*, 6:870, 2022.

- Robert Wille, Lukas Burgholzer, and Alwin Zulehner. Mapping quantum circuits to IBM QX architectures using the minimal number of SWAP and H operations. *Proc. DAC*, pages 1–6, 2019.
- Yong-Xin Yao, Niladri Gomes, Feng Zhang, Cai-Zhuang Wang, Kai-Ming Ho, Thomas Iadecola, and Peter P. Orth. Adaptive variational quantum dynamics simulations. *PRX Quantum*, 2(3):030307, 2021.
- Yordan S. Yordanov, Vasileios Armaos, Crispin H.W. Barnes, and David R.M. Arvidsson-Shukur. Qubit-excitation-based adaptive variational quantum eigensolver. *Communications Physics*, 4(1): 228, 2021.
- Xiao Yuan, Suguru Endo, Qi Zhao, Ying Li, and Simon C. Benjamin. Theory of variational quantum simulation. *Quantum*, 3:191, 2019.
- Alwin Zulehner, Alexandru Paler, and Robert Wille. An efficient methodology for mapping quantum circuits to the IBM QX architectures. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 38(7):1226–1236, 2018.